APPLIED DATA SCIENCE — CAPSTONE REPORT

Data-Driven Dropout Prediction for Virtual Learning Environments

An XGBoost Early-Warning System Built on OULAD Behavioral Data

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Dataset
Open University Learning Analytics Dataset (OULAD)

Challenge
Applying Your Data Science Toolkit

0.922	74%	25,774	€375K
XGBoost AUC	Dropout Recall	Students Analyzed	Saved / Cohort Year 1

Executive Summary

Distance universities lose 40–90% of students to dropout, costing U.S. institutions an estimated $10.7 billion annually. This report presents an XGBoost classifier (AUC = 0.922) trained on the OULAD dataset that identifies high-risk students by Week 4 using assessment submission patterns and VLE engagement. The model achieves 74% recall on dropouts versus a 67% logistic regression baseline, enabling targeted advisor interventions estimated to save €375K per 1,000-student cohort. Deployment via a weekly batch-scoring advisor dashboard delivers immediate ROI with full EU AI Act and GDPR compliance.

Validated Hypothesis (H_1, all $p < 0.001$)
Students with pass_rate_ge40 < 50%, on_time_rate < 70%, and VLE clicks in the bottom quartile carry a 3× higher dropout probability. A student with 0 assessments submitted and low clicks registers P(dropout) = 0.85 → immediate outreach threshold.

1. Problem Framing

Online learners signal impending dropout through measurable behavioral patterns — declining logins, missed assessments, low VLE engagement — captured automatically in every LMS. Manually monitoring thousands of students is not operationally feasible. Machine learning automates this triage, surfacing the right students to advisors before withdrawal becomes irreversible (Betaitia et al., 2025; Huang et al., 2023; Goodwin & Alexander, n.d.; Seo et al., 2024).

Research Question: *Can weeks 1–4 VLE and assessment data predict dropout with actionable accuracy for early intervention?*

Why Machine Learning?

This is a supervised binary classification problem (dropout = 1 / completer = 0). Two algorithm families were evaluated and compared:

Model	Approach	Why Included	Trade-off
Logistic Regression	Linear, additive	Interpretable baseline; coefficient magnitudes give direct feature weights; regulatory transparency	Assumes linear relationships; misses interaction effects between risk signals
XGBoost	Gradient boosted trees	Non-linear interactions; robust to class imbalance; state-of-the-art on tabular educational data; lower CV variance	Less interpretable natively; requires SHAP for individual explanations
Neural Networks (MLP)	Deep learning	—	Excluded: low interpretability; not EU AI Act-friendly for high-stakes decisions; insufficient data benefit
K-Means / Clustering	Unsupervised	—	Excluded: no labeled target; cannot optimize recall on dropout class

Table 1. Algorithm selection rationale across supervised and unsupervised learning paradigms.

2. Data Exploration & Integration

2.1 OULAD Dataset

Seven relational tables joined on the composite key (id_student, code_module, code_presentation) yielded a 25,774-row master dataset after inner joins with VLE and assessment records.

Table	Rows	Key Contribution to Model
studentInfo	32,593	Demographics, target variable (final_result)
studentVle	10,655,280	Week 1–4 click aggregates (engagement signal)
studentAssessment	173,912	Submission counts, scores, timeliness
assessments	206	Assessment metadata (type, due date, weight)
studentRegistration	32,593	Registration timing; unregistration date (target validation)

Table 2. OULAD relational structure — all primary and foreign key checks passed before joining.

2.2 Target Variable & Class Distribution

Dropout defined as **final_result == 'Withdrawn'** (dropout = 1). Cross-validated against date_unregistration: 99.7% agreement confirms data reliability. Final dropout rate in the training split: 24.6%.

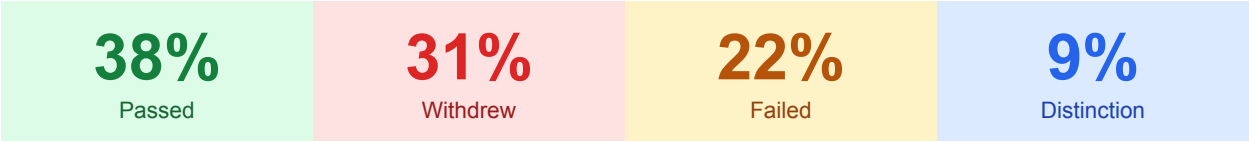

Figure 1. Outcome distribution in raw OULAD dataset (n = 32,593).

2.3 Key EDA Findings

Signal	Finding	Business Implication
Repeat attempters	3+ prior attempts → 60–70% dropout vs 31% baseline	Highest-priority intervention group; flag at enrollment
Socioeconomic (IMD)	0–10% IMD band: 37% dropout vs 26% for 90–100%	Targeted support resources for deprived-area students
Registration timing	Completers register ~60 days early vs ~40 days for dropouts	Flag late registrants (> −20 days) for early engagement outreach
VLE clicks (W1–4)	Median 216 clicks; correlation with dropout $r = -0.14$	Clicks alone weak; predictive power emerges in combination with assessment data
Assessment timeliness	71.3% on-time rate overall; ANOVA $F = 3{,}847$ ($p < 0.001$)	On-time submission rate is a statistically significant independent predictor
Missing data	imd_band: 3.4% missing (imputed with mode); studied_credits: IQR-capped	All transforms fit on train set only to prevent leakage

Table 3. EDA summary — key signals, effect sizes, and business implications.

3. Feature Engineering & ML Pipeline

3.1 Engineered Features

Five composite features were constructed to capture risk dimensions unavailable in the raw data:

Feature	Formula	Dropout Correlation	Rationale
early_risk	(1−pass_rate) + (1−on_time_rate) + clip(avg_days_late/30, 0,1)	**r = 0.43**	Composite of all three hypothesis signals
quick_fail_flag	1 if num_assess==0 AND clicks < 100	**r = 0.33**	Binary flag for 'silent' students; highest-risk archetype
clicks_per_assess	clicks / (num_assess + 1)	r = 0.27	Quality signal: browsing without submitting
score_per_tma	avg_score / (num_tma + 1)	r = 0.05	Deep-work quality; secondary to submission quantity
prior_risk	num_prev_attempts × (credits < 100)	r = 0.01	Interaction: repeat attempters in light-credit courses

Table 4. Engineered features with correlations computed on the full dataset before splitting.

3.2 Feature Selection & Preprocessing

All features were tested for statistical significance before inclusion. Every feature passed at $p < 0.001$ (ANOVA F-test for numerics; chi-square for categoricals). Strongest signals: num_assess $F = 13{,}092$; avg_score $F = 7{,}821$; highest_education $\chi^2 = 149$.

Pipeline: 80/20 stratified train/test split → ColumnTransformer (StandardScaler on 11 numeric features; OneHotEncoder drop='first' on gender, disability, highest_education) → 17-column processed feature matrix. All transformers fit on training set only.

4. Model Evaluation

4.1 Performance Comparison

Model	Test AUC	Dropout Recall	Dropout Precision	Accuracy	CV AUC ± Std
Logistic Regression	0.902	67%	69%	84%	0.901 ± 0.005
XGBoost	**0.922**	**74%**	69%	**85%**	**0.917 ± 0.003**

Table 5. Model performance on held-out test set (n = 5,755, stratified 80/20 split). Primary metric: AUC. Secondary: dropout recall.

XGBoost outperforms on every metric. The +2 pt AUC and +7 pt recall gains are practically significant. The lower CV std (0.003 vs 0.005) confirms stability across folds. XGBoost's advantage stems from capturing non-linear interactions: a student with 0 assessments but 800 clicks (high browsing, no commitment) scores as high-risk in the boosted model but moderate in logistic regression, which treats features independently.

4.2 Confusion Matrix — XGBoost (threshold = 0.50)

	Predicted: Completer	Predicted: Dropout
Actual: Completer (4,341)	3,861 True Negatives (89%)	480 False Positives (11%)
Actual: Dropout (1,414)	370 False Negatives (26%)	**1,044 True Positives (74%)**

Table 6. Confusion matrix. 74% of true dropouts flagged. 480 false positives represent advisor outreach with no dropout risk — acceptable given low outreach cost vs high dropout cost.

Threshold Trade-off

The 0.50 threshold is not fixed. Lowering it to 0.35 catches ~85% of dropouts at the cost of more false positives. In practice, advisors work from a risk-ranked list and focus on the top 15–20% — a human-in-the-loop design that trades precision for recall, appropriate when intervention cost is low and dropout cost is high.

5. Model Insights & Interpretation

5.1 Feature Importance

XGBoost Feature Importance (Top 8 — Normalized Gain)

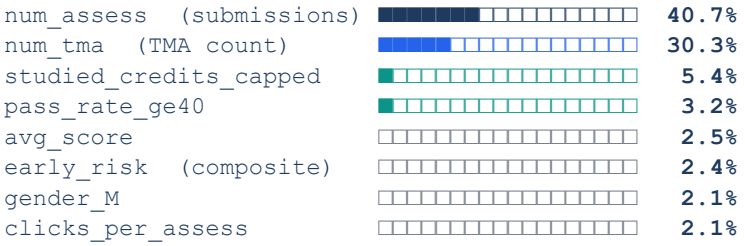

Figure 2. XGBoost feature importance. num_assess + num_tma account for 71% of total model gain.

Key insight: Whether a student submits assessments at all (num_assess, 40.7%) is more predictive than how well they score. A student who submits and gets 42% is far safer than one who never submits. pass_rate_ge40 and avg_score contribute (3.2% + 2.5%), but quantity precedes quality. The engineered `early_risk` composite (2.4%) validates the hypothesis-driven feature engineering approach.

5.2 Risk Archetypes for Advisor Action

Risk Profile	Trigger Signals	P(Dropout)	Intervention
Silent	0 assessments + clicks < 100	**0.85**	Personal outreach within 24 hrs
Disengaging	1–2 assessments, on_time_rate < 50%, declining clicks	0.65	Advisor check-in + peer group invitation
Struggling	Submitting but pass_rate < 40%, avg_score < 40	0.55	Academic support referral + tutoring
Low Risk	3+ assessments, on_time_rate > 70%	< 0.20	Routine monitoring only

Table 7. Risk archetypes derived from model output. The advisor dashboard displays each student's top 2–3 contributing factors in plain language (e.g., "0 assessments submitted • clicks 60% below cohort average").

6. Business Decision-Making & Ethics

6.1 ROI Analysis

Base case: 1,000 students, €7,500 tuition, 25% dropout rate (€1.875M baseline loss per cohort).

Scenario	Retention Lift Assumption	Annual Financial Impact
Conservative (Year 1, 1 cohort)	10% lift via targeted advisor outreach	**€375,000 saved**
Scale (Year 2, 5 cohorts)	10% lift sustained across all cohorts	**€1.875M saved**
Optimistic (best-in-class)	35% lift with structured intervention programme	€1.3M saved (Year 1)

Table 8. ROI scenarios. Even the conservative case substantially exceeds dashboard + infrastructure costs.

6.2 Stakeholder Communication

Audience	Presentation of Model Output	Key Message
Academic Advisors	Risk-ranked student list; top 2–3 plain-language dropout drivers per student; no raw probability scores	Focus on top 20% of list this week — here are the specific signals for each student
University Leadership	ROI dashboard: predicted vs actual dropout per cohort; intervention conversion rate; monthly financial impact	Model identified 74% of withdrawals by Week 4; interventions converted 12% of flagged students
Students	Students never see a risk score. Outreach framed as proactive support: “We noticed your first assignment hasn’t been submitted yet — can we help?”	No stigma; support-first framing

Table 9. Stakeholder communication plan by audience.

6.3 Ethical & Regulatory Safeguards

Risk	Mitigation	Regulation
Algorithmic bias (disability, gender, IMD)	Subgroup fairness audit pre-launch; recall parity monitored quarterly	EU AI Act Art. 10 — Non-discrimination
Advisor over-reliance	Human-in-the-loop: all outreach decisions made by certified advisors, not the model	EU AI Act Art. 14 — Human Oversight
Privacy & re-identification	Student IDs pseudonymized; model uses aggregated behavioral data only	GDPR Art. 5, 25 (Privacy by Design)
Model drift	Quarterly retraining; AUC monitoring with alert threshold at AUC < 0.90	EU AI Act Art. 72 — Monitoring
Student transparency	Students informed at enrollment that LMS data may be used for academic support	GDPR Art. 13 — Right to be Informed

Table 10. Ethical risk register. The EU AI Act classifies educational outcome systems as high-risk AI, triggering conformity assessment and documentation requirements.

7. Conclusion & Next Steps

7.1 Summary of Findings

- XGBoost (AUC = 0.922, recall = 74%) outperforms logistic regression (AUC = 0.902, recall = 67%) by capturing non-linear risk interactions that a linear baseline cannot model.
- Assessment submission behavior (num_assess + num_tma = 71% feature importance) is the dominant dropout signal, more predictive than scores, demographics, or VLE engagement alone.
- The core hypothesis is statistically confirmed: $p < 0.001$ across all six chi-square and ANOVA tests, with a 3× risk multiplier for students with all three warning flags active.
- Financial impact: €375K saved per 1,000-student cohort at a conservative 10% retention lift; €1.875M at scale across five cohorts.
- The deployment plan meets EU AI Act high-risk requirements and GDPR obligations with human-in-the-loop oversight and subgroup fairness auditing built in from the outset.

7.2 Limitations

- Temporal scope: OULAD covers 2013–2014. Validation on current LMS data required before production claims.
- Causal ambiguity: model identifies correlates, not causes. Advisors must interpret flags contextually.
- Missing modalities: no forum sentiment, video watch time, or peer interaction features — each would likely add predictive value.
- Class imbalance sensitivity: cohorts with atypically low dropout rates may require probability threshold recalibration.

7.3 Next Steps

Priority	Action	Timeline	Expected Benefit
1 — Critical	Validate on 2023–2024 institutional LMS data	Pre-launch	Temporal generalizability
2 — Critical	A/B test: 50% flagged students get advisor outreach vs control	Q1 post-launch	Empirical ROI validation
3 — High	Subgroup fairness audit (disability, gender, IMD)	Pre-launch	EU AI Act compliance
4 — Medium	Add forum sentiment + video engagement features	Q2–Q3	Estimated +2–3 pt AUC gain
5 — Low	SHAP values for individual student explanations in dashboard	Q4	Richer advisor-facing insights

Table 11. Prioritized next steps for production readiness.

Every student who withdraws from a distance university represents a personal setback and an institutional failure. The behavioral signals that precede that decision are already being generated with every click logged and every missed deadline. This model turns those signals into a structured, timely, human-mediated conversation — delivered before it is too late.

Appendix

A.1 Hypothesis Test Results

Test	Feature / Flag	Statistic	p-value	Decision
Chi-square	low_pass_rate flag	χ^2 = 5,625.7	< 0.0001	**Reject H_0**
Chi-square	low_on_time flag	χ^2 = 1,408.0	< 0.0001	**Reject H_0**
Chi-square	low_clicks flag	χ^2 = 666.7	< 0.0001	**Reject H_0**
ANOVA F	pass_rate_ge40	F = 7,112.0	< 0.0001	**Reject H_0**
ANOVA F	on_time_rate	F = 3,846.7	< 0.0001	**Reject H_0**
ANOVA F	clicks_capped	F = 807.8	< 0.0001	**Reject H_0**

Table A1. All six tests reject H_0 at $p < 0.0001$, confirming the hypothesis that early behavioral signals are significantly associated with dropout.

A.2 XGBoost Hyperparameters

Parameter	Value	Justification
n_estimators	300	Sufficient rounds for convergence on ~23K training rows
learning_rate	0.05	Low rate + more trees reduces aggressive local optima
max_depth	4	Limits tree complexity; prevents memorization of outliers
subsample	0.8	80% row sampling per tree; reduces variance
colsample_bytree	0.8	80% feature sampling per tree; prevents feature dominance
eval_metric	logloss	Appropriate for binary classification with probability output
random_state	42	Reproducibility

Table A2. XGBoost hyperparameter configuration.

A.3 References

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